MODULE 08 · SECTION 1 - FUNDAMENTALS OF GEN AI

Google's foundation models: Gemini, Gemma, Imagen, Veo

Four families named in the guide, what each is for, and what has changed in 2026.

SECTION 1 of 4 EXAM WEIGHT ~30% MODULE 8 of 33

admin@sthithapragna.com

Four families, four jobs. Gemini **does almost everything**; the other three are specialists.

The four families at a glance

FAMILY	WHAT IT IS	CHOOSE IT WHEN
Gemini	Multimodal all-rounder	Text, images, code, many tasks
Gemma	Open, lightweight models	You must run it yourself
Imagen	Text-to-image	You need images from text
Veo	Text or image to video	You need video clips

THE TIEBREAKER

Broad, mixed tasks point to Gemini. One narrow output, or an open-model constraint, points to a specialist.

Gemini: the multimodal all-rounder

- **Multimodal:** understands text, images, audio, video and code together.
- **Long context:** many Gemini models take 1 million tokens or more.
- **Reasoning:** thinking models plan and work through multi-step problems.
- **Generates** text and code, plus images through Gemini's own image models.
- **Everywhere:** powers the Gemini app, Workspace, Gemini Enterprise and Agent Platform.

> ON THE EXAM
>
> Versatile, multimodal, long documents, mixed tasks for teams without AI expertise: Gemini.

Gemini comes in tiers

TIER	BUILT FOR
Pro	The hardest reasoning, coding and analysis
Flash	The everyday workhorse: strong quality at speed
Flash-Lite	The lowest cost, for huge volumes of simple tasks
Live	Real-time voice conversations with low delay
Image models	Generating and editing images, nicknamed Nano Banana

REMEMBER

Version numbers change every few months. Learn the tiers and what each is for, not the numbers.

Gemma: open models you can run yourself

- **Open models**, built from the same research and technology as Gemini.
- **Lightweight:** run on your own hardware, mobile devices or a hosted service.
- **Control:** download, fine-tune and keep data on your own systems.
- **Variants:** Gemma 4, Gemma 3n for devices, CodeGemma, MedGemma, ShieldGemma.
- **Trade-off:** smaller than Gemini, and you run and maintain it.

> ON THE EXAM
>
> Open, lightweight, local, fine-tune on your own hardware: Gemma.

Gemini or Gemma?

GEMINI

Managed and most capable

- Google runs it for you
- Largest, most capable models
- Multimodal at enterprise scale
- Use through apps or an API

GEMMA

Open and in your hands

- You download and run it
- Light enough for modest hardware
- Data can stay on your machines
- Fine-tune and adapt freely

THE TIEBREAKER

Need it open, local or on limited hardware? Gemma. Need the most capable managed model? Gemini.

Imagen: images from text

- **Imagen** generates high-quality images from a text description.
- **Uses:** campaign visuals, product mock-ups, illustrations, concept art.
- **Currency check:** every Imagen model shut down on 17 August 2026.
- **The exam guide still lists it** as one of Google's four foundation models.
- **On the exam:** if an option says Imagen for text-to-image, it can be right.

> ON THE EXAM
>
> Answer from the guide's world. Imagen for images from text is still the examinable answer.

What replaced Imagen: Gemini's image models

- **Gemini image models**, nicknamed Nano Banana, now handle image generation.
- **Conversational editing:** change an image by asking, turn by turn.
- **Combine images** and keep the same character or product consistent.
- **Legible text** inside images, such as labels and signs.
- **Google's migration guide** maps each Imagen model to a Gemini image model.

> REMEMBER
>
> In real projects today, image generation means a Gemini image model. On the exam, Imagen remains valid.

Veo: video from text and images

- **Veo** generates video from a text prompt, or from reference images.
- **Veo 3.1** also generates audio natively: sound effects, ambience, dialogue.
- **Uses:** storyboards, short adverts, product clips, previews of a scene.
- **Watermarked:** Google marks generated video with an invisible SynthID watermark.
- **Next:** Gemini Omni, in Preview, also generates and edits video.

> THE TIEBREAKER
>
> Moving pictures from a description or stills: Veo. Still pictures: Imagen or Gemini's image models.

Other Google model names you may see

NAME	WHAT IT DOES
Chirp	Speech models: natural voices and speech recognition
Lyria	Generates music from text and image prompts
Embeddings models	Turn text or images into numbers for meaning-based search
Gemini Live	Real-time, two-way voice conversation
MedGemma, CodeGemma	Gemma variants for healthcare and for code

ON THE EXAM

These appear as wrong options. Know what each does so you can rule it out quickly.

Match the job to the model

THE JOB	THE MODEL
Draft emails, summarise reports, help with code	Gemini
A chatbot that runs offline on a laptop	Gemma
A campaign image from a written brief	Imagen, or a Gemini image model
A 15-second advert from a script	Veo
Spoken audio from a written script	Chirp voices in Text-to-Speech
A jingle for the advert	Lyria

THE TIEBREAKER

Name the output first: text, image, video, speech or music. The family usually follows.

Wrenfield's model choices

WRENFIELD NEED	MODEL FAMILY
Product copy and the Dutch translations	Gemini, a fast tier such as Flash
Find themes across 18 months of call transcripts	Gemini, for its long context
Shop-floor tablets that must work offline	Gemma, running on the device
Lifestyle images for the autumn campaign	A Gemini image model
A 15-second sofa advert for social media	Veo

THE TAKEAWAY

One company, four families. Choosing a model is choosing per job, not once for the whole business.

> Wrenfield's contact centre receives photos of damaged furniture together with a customer's message. Agents want a draft reply that reflects what the photo shows.
>
> The team wants a fully managed service, with no models or servers to run.

Which model family fits best?

A Gemini

B Gemma

C Imagen

D Veo

◆ PAUSE AND DECIDE

Which model family fits best?

A Gemini

Reads the photo and message together, then drafts text. Fully managed.

B Gemma

Can read images, but you run it. They want it managed.

C Imagen

Creates images. It does not read them or write replies.

D Veo

Creates video. Nothing here needs a video.

THE SIGNAL

Understand an image, write text, fully managed: Gemini.

> An estate agency wants short promotional clips for new listings. Staff will provide a written description and a few still photos of each home.
>
> Each clip should be around 20 seconds, with background sound.

Which model family should the agency use?

A Imagen

B Gemma

C Chirp

D Veo

◆ PAUSE AND DECIDE

Which model family should the agency use?

A Imagen

Creates still images, not clips.

B Gemma

An open language model. It does not generate video.

C Chirp

Speech only. It could voice a clip, not create one.

D Veo

Video from text and reference images, with native audio.

THE SIGNAL

The output is a moving clip, made from text and stills: Veo.

> A hospital research unit wants a model to summarise clinical notes. Policy forbids sending patient data to any external API, so everything must run on the hospital's own servers.
>
> The team wants a lightweight, open model it can fine-tune on its own data.

Which Google model family fits?

A Gemma, run on the hospital's own servers

B Gemini, called through its API

C Veo, with a summary prompt

D Imagen, fine-tuned on the notes

◆ PAUSE AND DECIDE

Which Google model family fits?

A Gemma, run on the hospital's own servers
Open, lightweight, runs locally, and can be fine-tuned.

B Gemini, called through its API
Capable, but the policy forbids external APIs.

C Veo, with a summary prompt
Veo makes video, not text summaries.

D Imagen, fine-tuned on the notes
Imagen makes images, and is now retired.

THE SIGNAL

Must stay on our servers, open, lightweight, fine-tunable: Gemma.

> It is October 2026. Wrenfield's marketing plan, written in 2025, says product images will be generated with Imagen.
>
> The developer starting the project asks what to use now.

What is the best advice?

A Keep the plan; Imagen is still the latest image model

B Switch to Veo and take still frames from the video

C Use Gemma, since it is open and free to adapt

D Use a Gemini image model, which replaced Imagen

◆ PAUSE AND DECIDE

What is the best advice?

A Keep the plan; Imagen is still the latest image model

Imagen shut down on 17 August 2026.

B Switch to Veo and take still frames from the video

A clumsy workaround. Veo is built for video.

C Use Gemma, since it is open and free to adapt

Gemma generates text, not images.

D Use a Gemini image model, which replaced Imagen

Google's migration guide maps Imagen to Gemini image models.

THE SIGNAL

A real project in late 2026 uses a Gemini image model. The exam may still say Imagen.

> This one is written in the exam's own style. A marketing team needs high-quality images generated from text descriptions for a seasonal campaign.
>
> They want a Google foundation model named in the exam guide.

Which foundation model should they use?

A Veo

B Gemma

C Imagen

D Chirp

◆ PAUSE AND DECIDE

Which foundation model should they use?

A **Veo**
Video, not still images.

B **Gemma**
An open language model. No image generation.

C **Imagen**
The guide's text-to-image model. The examinable answer.

D **Chirp**
Speech, not images.

THE SIGNAL

Images from text, guide-named model, no Gemini image option: Imagen.

> A procurement team must answer questions about a 1,500-page tender pack. Answers often depend on comparing clauses that are hundreds of pages apart.
>
> They want the whole pack considered in a single request.

Which option fits best?

A Gemma 3n running on staff phones

B A Gemini model with a long context window

C Gemini Flash-Lite, sent one page at a time

D A small open model fine-tuned on old tenders

◆ PAUSE AND DECIDE

Which option fits best?

A Gemma 3n running on staff phones

Built for small devices, not a 1,500-page pack in one go.

B A Gemini model with a long context window

Holds the whole pack at once, so distant clauses can be compared.

C Gemini Flash-Lite, sent one page at a time

Page by page, it can never compare clauses far apart.

D A small open model fine-tuned on old tenders

Old tenders do not teach it this one, and it will not fit.

THE SIGNAL

The whole document in one request, with cross-references: long context.

> Wrenfield must translate 48,000 short product descriptions into Dutch before launch. Quality must be good, but this is simple, repetitive work.
>
> The finance team wants the lowest cost that still does the job well.

Which Gemini tier is the sensible starting point?

A. A fast, low-cost tier such as Flash or Flash-Lite

B. The Pro tier with a high thinking level

C. Gemini Live

D. A newly pre-trained translation model

◆ PAUSE AND DECIDE

Which Gemini tier is the sensible starting point?

A. A fast, low-cost tier such as Flash or Flash-Lite
Built for high-volume, simpler tasks at low cost.

B. The Pro tier with a high thinking level
Pays for reasoning the task does not need.

C. Gemini Live
For real-time voice conversation, not batch translation.

D. A newly pre-trained translation model
Months of cost for a job existing models already do.

THE SIGNAL

High volume, simple task, lowest cost that works: start with a fast tier, test quality, step up only if needed.

> A museum wants spoken audio guides for each gallery, in five languages, generated from scripts its curators have already written.
>
> The voices must sound natural.

Which Google offering fits best?

A Lyria

B Veo

C Imagen

D Text-to-Speech with Chirp voices

◆ PAUSE AND DECIDE

Which Google offering fits best?

A Lyria

Generates music, not narration.

B Veo

Generates video. Overkill, and no gallery video is wanted.

C Imagen

Generates images.

D Text-to-Speech with Chirp voices

Turns written scripts into natural speech, in many languages.

THE SIGNAL

Written script in, natural speech out: Text-to-Speech.

Module 8 in five lines

- Gemini: the managed, multimodal all-rounder, in Pro, Flash and Flash-Lite tiers.
- Gemma: open, lightweight models you run and fine-tune yourself.
- Imagen: text to image. Retired in August 2026, still examinable.
- Veo: video from text or images, now with native audio.
- Name the output first; then match the family and the tier.